

Pranshu Nijhawan

Principal Architect - Agentic AI & Platform Architecture

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Summary

Principal Architect with 10+ years in enterprise SaaS. Accountable for the architecture of **Promotion Advisor**, a **multi-tenant SaaS** in McKinsey's Periscope suite, covering its planning application, Analytics Engine, and agentic capabilities. Recent work spans agentic architecture across an **MCP mesh** (tool discovery and routing, deterministic business logic, agent data-access control) and high-performance data systems (sub-2 second response on 2M+ records, billion-row analytics on ClickHouse, semantic vectorized lookup over JSONB). Coached org-wide **Cursor** and AI-assisted engineering practices for a 37% capacity increase, and led the shift to **Spec-Driven Development** and an **AI-led SDLC**.

Experience

McKinsey & Company

March 2022 - Present

Principal Architect

Senior Software Engineer (2022) · Senior Software Engineer II (2023) · Principal Architect (2025)

Own the architecture of [Promotion Advisor](#), a **multi-tenant SaaS** in McKinsey's [Periscope](#) suite, covering its planning application, Analytics Engine, and agentic layer. Co-built **Platform AI**, the agent platform every Periscope product deploys through, and wrote the suite's first unified architecture reference. The cost and API standards it set now govern eight sibling products.

- **Platform Architecture Ownership:** Cut platform infrastructure spend 40% through the cost standards I set. Own technical direction, ADRs, and release architecture across five squads, covering the planning application, Analytics Engine, agentic layer, configuration framework, security model, and external API surface.
- **Platform AI: Enterprise Agent Platform:** Co-built **Platform AI**, Periscope's enterprise agent platform: a GitOps **Agent Definition Manager** that versions every agent as Git artifacts, not prompt text; a LangChain harness on Kubernetes running one agent loop for all product agents; and a Tenant MCP gateway owning authentication, tool discovery, and quota. Caller identity survives every hop.
- **Cortex: Deterministic Business-Logic Engine:** Built **Cortex**, a stateless engine that runs business logic written as configuration. Rules compose from three operations (acquire, derive, assert) into playbooks; a purpose-built TypeScript compiler parses the typed YAML into executable binaries. Business rules stay deterministic and unit-testable, so agent output is reproducible and auditable.
- **Context-Scoped Catalog: Tool Selection at Fleet Scale:** No version-mismatch incident in 6 months, and a new solution now registers without per-tool wiring. Designed a tiered selection algorithm (pinned tools, a BM25 shortlist, then a cheap router model for near-duplicates) and scoped the catalog per tenant, keeping agent context bounded as the tool estate grows.
- **Periscope Reporting Engine on ClickHouse:** Removed the batch-prep step and the analytics-team dependency for every new measure: users compose their own reports and each aggregation resolves at request time against a governed ontology through a dynamic Query Builder. Built on ClickHouse with billion-row sub-second queries.
- **Real-Time Promotion Conflict Detection:** Sub-2s conflict detection at 20 concurrent users across 2M promotions. Designed the multi-dimensional algorithm: progressive candidate narrowing over sorted item arrays, which keeps the check interactive as promotion volume grows.
- **AI-Assisted Engineering & Agent Tooling:** Drove org-wide adoption of **Cursor** and **Spec-Driven Development** (OpenSpec, SpecKit), which raised team capacity 37%. Built the agent library behind it: 10+ production agents for architecture diagrams, ADR and APR authorship, and technical communications, plus an SDD Agent that runs an 11-dimension anti-hallucination audit per spec. The **Repo Mastery Agent** replaced a heavy Graph RAG documentation system with a Registry Pattern over vector binaries and explicit dependency relations.

Technologies: MCP · LangChain · Cursor · TypeScript · Python · Node.js · C# · .NET · React · PostgreSQL · ClickHouse · Databricks · Kubernetes · Azure.

Eptura (formerly Condeco)

April 2021 - March 2022

Principal Engineer

Returned as Principal Engineer; previously Software Engineer, 2018 - 2019

Re-architected a single-tenant workspace SaaS to cloud-native microservices on AKS and built a global IoT release orchestrator that pushes firmware to 5M+ devices. Directed 20 engineers across four squads.

- **Cloud-Native Re-Architecture:** Migrated legacy single-tenant SaaS to cloud-native microservices on AKS (Azure Kubernetes Service): 60% faster incident recovery, eliminated single-tenant scaling bottlenecks.
- **Global IoT Release Orchestration:** Built a release orchestration system (Azure IoT Hub, Azure Functions, Azure CosmosDB) that delivers firmware updates to 5M+ IoT devices worldwide with zero-downtime rollouts.
- **Engineering Leadership:** Directed 20 engineers (4 tech leads) across four squads. Established a structured mentorship program: +20% retention, accelerated engineering progression.

Technologies: C# · .NET · React · Angular · AKS · Azure IoT Hub · Azure Functions · Azure CosmosDB · Azure · SQL Server

Nagarro

Senior Engineer

December 2019 - April 2021

Shipped full-stack web and mobile apps for e-commerce and HR enterprise clients on Angular, .NET Core, PostgreSQL, and Ionic. CI/CD and IaC practices cut deployment failures by 70% across client projects.

- **Full-Stack Web + Mobile Delivery:** Delivered production web and mobile apps on Angular, .NET Core, PostgreSQL, and Ionic Framework at 99.5%+ uptime.
- **CI/CD & Infrastructure as Code:** Established CI/CD pipelines and IaC practices: 70% reduction in deployment failures across client projects.

Technologies: C# · .NET Core · Angular · Ionic Framework · PostgreSQL · DynamoDB · AWS · Fargate

Eptura (formerly Condeco)

Software Engineer

September 2018 - December 2019

Built C# / .NET / React / SQL Server microservices that carried 3x customer growth without reliability regressions. Cut build times 40% and core query latency 3x through data model refactoring.

- **Microservices for 3x Customer Scale:** Designed and shipped C# / .NET / React / SQL Server microservices that carried 3x customer growth without reliability regressions.
- **CI/CD & Data Model Optimization:** Cut build times by 40%; refactored core data models to reduce average query latency by 3x.

Technologies: C# · .NET · React · Angular · SQL Server · Azure

Gartner

Associate Software Engineer

February 2016 - August 2018

Built backend features on C# / .NET Core / React for an HR analytics SaaS, which improved survey data accuracy 60% for consulting clients. Automated SQL workflows saved 240+ engineering hours annually.

- **SaaS Analytics Backend:** Shipped backend features on C# / .NET Core / React: improved survey data accuracy by 60% for HR consulting clients.
- **SQL Workflow Automation:** Automated SQL workflows (stored procedures, SQL Agent jobs): saved 240+ engineering hours annually.

Technologies: C# · .NET Core · React · SQL Server · AWS

Core Competencies

- **AI & Agentic Engineering:** Agentic AI Architecture, Multi-Agent Orchestration, MCP (servers and clients), Tool Discovery and Routing, Context Engineering, Deterministic Business-Logic Compilation, Agent Data-Access Control, RAG (Retrieval-Augmented Generation), Cursor (Rules, Skills, Agents, Hooks, SDK), Spec-Driven Development (OpenSpec, SpecKit, BMad).
- **SaaS & Architecture:** System Design, Microservices, Distributed Systems, Domain-Driven Design (DDD), API Design (REST, GraphQL), Configuration-Driven Architecture, Polyglot Persistence, Multi-Tenant SaaS, ADR / APR Authorship.
- **Cloud & Infrastructure:** Azure, AWS, Kubernetes (AKS), Terraform, GitOps, CI/CD, Zero-Downtime Delivery, Observability (Dynatrace, OpenTelemetry).
- **Data Engineering & Real-Time Analytics:** Databricks, Spark (PySpark), DBT, ClickHouse, PostgreSQL, Redis, Real-Time Analytics, ELT / ETL, Columnar Analytics, FinOps / Cost Optimization.
- **Tech Stack:** TypeScript, Python, C#, .NET Core, Node.js, Fastify, React, Angular, SQL.

Educational Qualifications

Gurgaon Institute of Technology & Management

2012 - 2016

Bachelor of Technology (B. Tech), Computer Science & Engineering.